

Submission Summary

Conference Name	6th IEEE India Council International Subsections Conference
Paper ID	2273
Paper Title	Shoplifting Detection Using Convolutional LSTM Networks with Adam Optimizer
Abstract	Retail theft, particularly shoplifting, remains a critical issue for businesses, leading to considerable financial losses each year. This research introduces an advanced framework for real-time shoplifting detection using surveillance systems. The approach leverages a hybrid Convolutional Long-Short-Term Memory (ConvLSTM) network, effectively combining spatial and temporal data analysis for enhanced anomaly detection. The convolutional layers identify the spatial characteristics of individual video frames, while the LSTM components analyze sequential dependencies, enabling the recognition of intricate shoplifting behaviors. To optimize training efficiency and accelerate convergence, the Adam optimizer is employed, which dynamically fine-tunes learning rates based on gradient estimates. The underlying mathematical principles of this optimization technique are explored, demonstrating its effectiveness in stabilizing learning. Extensive tests were performed using a diverse dataset of shoplifting scenarios, where the proposed model achieved classification accuracy 83%, surpassing conventional techniques. In addition, the study discusses challenges such as dataset imbalances and environmental factors that may affect the accuracy of detection. Future enhancements, including sophisticated augmentation strategies and attention-based mechanisms, are suggested to improve the adaptability and precision of the model. This work contributes to the development of intelligent, scalable and automated theft detection solutions for modern retail security systems. Index Terms—Shoplifting detection, video surveillance, deep learning, anomaly detection, ConvLSTM, retail security, realtime monitoring, Adam optimizer.
Created	4/28/2025, 4:00:22 PM
Last Modified	4/28/2025, 4:06:05 PM
Authors	Gargi Singh (KIET Group Of Institutions) <singhgargi077@gmail.com>
Primary Subject Area	AI & ML, Data Science
Submission Files	Shoplifting.pdf (1 Mb, 4/28/2025, 4:00:06 PM)